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SUPERSEDING
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MILITARY SPECIFICATION

VOICEMITTER-OUTLET VALVE, C29R1; COMPONENTS FOR

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the components for the C29R1 voicemitter-outlet valve.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

DRAWINGS

EDGEWOOD ARSENAL

DL5-81-3 — Voicemitter - Outlet Valve, C29R1 — Data List. (All drawings referenced in this specification are listed on Drawing DL5-81-3.)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies of these freight classification rules should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Materials and components. All materials and components cited on Drawing DL5-81-3, or on the subsidiary drawings,

FSC 1240

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shall conform to the specifications thereon or to the specific characteristics set forth on the drawings.

3.2 **Manufacture and assembly.** The components shall be manufactured and assembled as specified on the drawings which are listed on Drawing DL5-81-3.

3.3 **Valve seat leakage.** Leakage through the valve seat shall not exceed 15 milliliters of air per minute at a pressure differential of 25 millimeters of water when tested as specified in 4.7.1.

3.4 **Frame-retaining ring and outlet-valve seat assembly.** The crimped joint of the frame-retaining ring and the cemented joint between the frame and outlet-valve seat assembly shall not leak when tested as specified in 4.7.2. The preproduction sample shall not leak when tested as specified in 4.7.3.

3.5 **Impact resistance.** The outlet valve seat shall not break nor shall the ribs or hub be depressed so that the distance from any portion of the upper surface of the hub to the upper edge of the sealing-rim surface (see figure 1) has been increased by more than 0.012 inches, when tested as specified in 4.7.4.

3.6 **Preproduction sample.** Prior to the start of regular production, a preproduction sample of 5 of each component, except those components covered by separate specifications, shall be produced using the same methods, materials, and equipment as will be used during regular production.

3.7 **Workmanship.** The voicemitter-outlet valve components shall be free from imperfections, foreign matter, and damage. Imperfections shall include blisters, porosity, cracks, tears, and excessive flash for the rubber components, and dents, cracks, burrs, sharp edges, surface roughness, and distortion for the metal components. Damage shall constitute tears, cracks, dents, and distortion as applicable.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure that supplies and services conform to specified requirements.

4.2 **Special provisions.**

4.2.1 *Alternative inspection.* The supplier may utilize any alternative inspection procedure which will provide equal or better assurance of quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 *Objective evidence.* The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2.3 *Inspection substations.* If the supplier elects to establish inspection substations for the purpose of conducting inspection for certain listed defects, the Acceptable Quality Levels (AQL's) for each inspection substation shall be equal to the AQL which is the next numerically lower value than the AQL specified for the total listed defects. The requirements of MIL-STD-105 are applicable to each substation and to the end item.

4.3 **Preproduction sample inspection.**

4.3.1 *Examination.* The sample compo-

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nents shall be inspected for all requirements of the drawings and specifications.

4.3.2 *Tests.* The sample compoents shall be tested in accordance with 4.7.

4.3.3 *Acceptance/rejection criteria.* The preproduction sample of each component shall meet the examination and tests specified by 4.3.1 and 4.3.2 to be acceptable. The supplier shall obtain written approval from the contracting officer before proceeding with regular production.

4.4 **Lotting.** A regular production lot shall consist of like components produced by one manufacturer from the same materials, at the same plant, and under essentially the same manufacturing conditions.

4.5 **Sampling.**

4.5.1 *For examination and nondestructive tests.* Sampling shall be conducted in accordance with MIL–STD–105.

4.5.2 *For destructive tests.* Sampling shall be conducted in accordance with MIL–STD–105 using level S–4.

4.6 **Inspection provisions.**

4.6.1 *For examination and nondestructive tests.* Sample components from the regular production lots shall be examined in accordance with the classification of defects and with MIL–STD–105

4.6.2 *For destructive tests.* The sample taken in accordance with 4.5.2 shall be tested in accordance with 4.7.4 using an AQL of 0.65 percent defective.

4.6.3 *Classification of defects.*

4.6.3.1 *Ring, sealing (Dwg B5–81–4).*

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Diameter (2.425) incorrect
102	Dept (0.235) incorrect
*103	Radii incorrect
104	Workmanship (3.7)
Minor:	AQL 2.5 percent defective
*201	Diameter (2.475) incorrect
*202	Depth (0.105) incorrect
*203	Gap (5/64) incorrect
204	Protective finish incorrect or missing
*205	Diameter (2.015) incorrect
*206	Diameter (2.490) incorrect

* Tool controlled (may be periodically inspected).

4.6.3.2 *Gasket (Dwg B5–81–6).*

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Workmanship (3.7)
*102	Outside diameter incorrect
103	Thickness incorrect
104	Foreign matter (grease or oil)

* Tool controlled (may be periodically inspected).

4.6.3.3 *Baffle, front (Dwg C5–81–8).*

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Outside diameter incorrect
102	Depth (0.060) incorrect
103	Workmanship (3.7)
Minor:	AQL 2.5 percent defective
*201	Cutouts incorrect or incorrectly located
202	Protective finish incorrect or missing

* Tool controlled (may be periodically inspected).

4.6.3.4 *Baffle, cupped (Dwg C5–81–9).*

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Diameter (2.315) incorrect
102	Diameter (2.150) incorrect
103	Flange height incorrect
104	Workmanship (3.7)
Minor:	AQL 2.5 percent defective
201	Depth (0.145) incorrect
202	Cutouts incorrect or incorrectly located

* Tool controlled (may be periodically inspected).

4.6.3.5 *Cup, inner (Dwg B5–81–10).*

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
*101	Outside diameter incorrect
*102	Inside diameter incorrect

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Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
103	Flange height incorrect
104	Workmanship (3.7)
Minor:	AQL 2.5 percent defective
201	Radius incorrect

• Tool controlled (may be periodically inspected).

4.6.3.6 *Baffle, inner (Dwg C5–81–11).*

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Outside diameter incorrect
102	Projection depth (0.062) or radius incorrect
103	Depth (0.020) incorrect
104	Workmanship (3.7)
Minor:	AQL 2.5 percent defective
*201	Diameter (2.031) incorrect
*202	Location of Projections incorrect
203	Cutouts incorrect or incorrectly located
204	Protective finish incorrect or missing

• Tool controlled (may be periodically inspected).

4.6.3.7 *Gasket, baffle (Dwg B5–81–12).*

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Workmanship (3.7)
*102	Outside diameter incorrect
*103	Inside diameter incorrect
104	Thickness incorrect
105	Foreign matter (grease or oil)

• Tool controlled (may be periodically inspected).

4.6.3.8 *Frame and retaining-ring assembly (Dwg C5–81–14).*

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 1.0 percent defective	
101	Component missing or incorrectly assembled	
≠102	Plain view profile incorrect	
103	Frame-retaining ring crimped joint defective or incorrect	
¢104	Diameter (2.417) incorrect	
¢105	Diameter (2.465) incorrect	
¢106	Diameter (2.367) incorrect	
¢107	Depth (0.178) incorrect	
≠108	Diameter (1.498) incorrect	
109	Workmanship (3.7)	
150	Valve seat leakage	4.7.1
Major A:	AQL 0.25 percent defective	
151	Frame-retaining ring and outlet valve seat assembly leakage	4.7.2
Minor:	AQL 2.5 percent defective	
¢*201	Diameter (2.064) incorrect	
¢*202	Depth (0.354) incorrect	
¢*203	Height (0.095) incorrect	
≠*204	Diameter (1.515) incorrect	
≠*205	Diameter (1.255) incorrect	
≠*206	Depth ($1\frac{1}{16}$) incorrect	
≠*207	Diameter (2.070) incorrect	
208	Radii or retaining beads missing	
209	Protective finish missing or incorrect	

≠ Reference Dwg C5-81-7.

¢ Reference Dwg B5-81-5.

• Tool controlled (may be periodically inspected).

4.6.3.9 *Seat, outlet valve (Dwg C5–81–16).*

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 1.0 percent defective	
101	Workmanship (3.7)	
102	Diameter (1.130) incorrect	
103	Diameter (1.512) incorrect	
104	Diameter (0.176) incorrect	
105	Thickness (0.150) incorrect	
106	Thickness (0.160) incorrect	
107	Sealing rim surface not smooth	
108	Countersink diameter (0.195) incorrect	
*Major A:	AQL 2.5 percent defective	
150	Valve seat leakage	4.7.1
*Major:	AQL 2.5 percent defective	
2[illegible]	[illegible] or illegible	

• For prime procurement only

4.7 Tests.

4.7.1 *Valve seat leakage.* An outlet valve disc (Dwg B5–2–1006) shall be assembled in the frame, in accordance with Drawing D5–81–3, and tested using Edgewood Arsenal M4A1 Outlet Valve Leakage Indicator (see 6.3).

4.7.2 *Frame-retaining ring and outlet-*

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valve seat-assembly leakage. The leakage of the frame-retaining ring and outlet-valve seat assembly srall be determined using the Edgewood Arsenal Q25 Air Pressure Test Device, at a static pressure of 3 pounds per square inch, for 15 seconds.

4.7.3 *Preproduction sample.* Repeat the leakage test specified in 4.7.2 on the preproduction sample after the assemblies have been subjected to a temperature of —40° ± 2°F. for 24 hours and again after they have been subjected to boiling water for two hours.

4.7.4 *Impact resistance. Impact* resistance of the outlet valve seat shall be determined using the Edgewood Arsenal Q26 Impact and Rigidity Mechanical Tester, with a ¼-pound weight dropped from a height of 24 inches.

5. PREPARATION FOR DELIVERY (For repair parts packaging see 6.4)

5.1 Packaging, level C. Components of like description shall be packaged to afford adequate protection from the supply source to the first receiving activity for immediate use.

5.2 Packing, level C. Like components, packaged as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall comply with Uniform Freight Classification ratings, rules, and regulations of other carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit and intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The components covered by this specification are intended to be used in the manufacture of the C29R1 voicemitter-outlet valve.

6.2 Ordering data. Procurement documents should specify the title, number, and date of this specification.

6.3 Outlet valve disc. The outlet valve discs required for use in performing the valve seat leakage test will be furnished by the Government and should be obtained as directed by the contracting officer.

6.4 Repair parts packaging. When repair parts packaging and packing is required for items authorized for stocking as repair parts, the repair parts packaging data sheets covering the compoents for the C29R1 Voicemitter-Outlet Valve should be used. (Copies of repair parts packaging data sheets may be obtained from the Director of Engineering & Industrial Services, ATTN: Chief, Design Branch, US Army Edgewood Arsenal, Md. 21010.)

6.5 International interest. Certain provisions of this specification are the subject of ABC international standardization agreement. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

Custodians:
Army—MU
Navy—SH
Air Force—84

Preparing activity:
Army—MU(EA)
Project No. 1240-0077

International interest (see 6.5)

Review activities:
Army—MD
Navy—SH

User activities:
Navy—MC
Air Force—84

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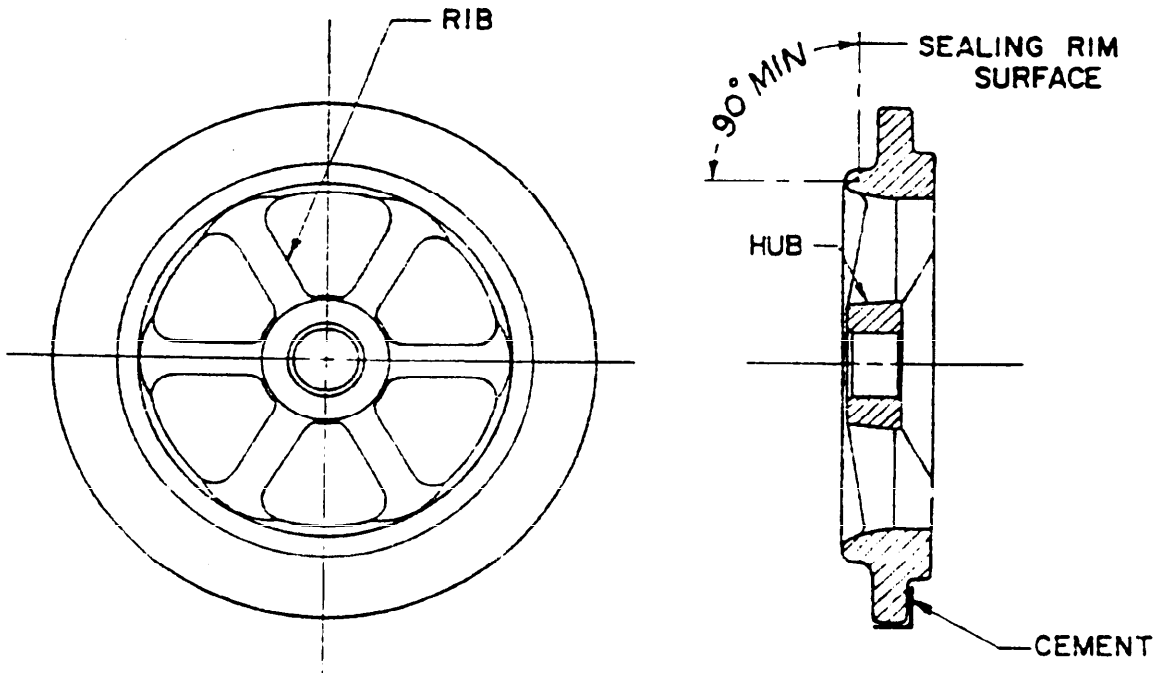


FIGURE 1. *Outlet valve seat.*

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